Smoothed Wasserstein Error for Langevin Diffusion Sampling

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What is sampling?

- We have a target distribution π on $\mathbb{R}^d$.
- **Sampling** means generating a random variable X such that

$$X \sim \pi.$$

- We care because we want expectations (integrals):

$$\mathbb{E}_\pi[f(X)] = \int_{\mathbb{R}^d} f(x)\,\pi(x)\,dx.$$

- Sampling turns hard integrals into averages (Monte Carlo):

$$\mathbb{E}_\pi[f(X)] \approx \frac{1}{n}\sum_{i=1}^{n} f(X_i), \qquad X_i \sim \pi.$$

Purpose: sampling is a computational tool to approximate integrals and quantify uncertainty.

Why can't we just sample directly?

In many problems, the density is known only up to a constant:

$$\pi(x) \propto e^{-V(x)}.$$

That is,

$$\pi(x) = \frac{e^{-V(x)}}{Z}, \qquad Z = \int_{\mathbb{R}^d} e^{-V(x)}\, dx.$$

- The normalizing constant Z is usually **intractable** in high dimension.
- Inverse-CDF sampling is typically impossible (no closed-form CDF; high-d).
- Rejection sampling often becomes **exponentially inefficient** as d grows.

Core idea: instead of sampling directly, build a stochastic process whose long-run distribution is π.

Why Langevin SDE? (intuition)

We want samples to spend most of their time where π is large.
If $\pi(x) \propto e^{-V(x)}$:

- Low $V(x) \Rightarrow$ high probability (typical / likely region).
- High $V(x) \Rightarrow$ low probability (rare region).

So we want dynamics that:

1. drift **downhill** in V (toward likely regions),
2. but still **explore** (not collapse to a minimizer).

The Langevin SDE does exactly that:

$$dX_t = -\nabla V(X_t)\, dt + \sqrt{2}\, dB_t.$$

- $-\nabla V(X_t)\, dt$: gradient flow toward low energy.
- $\sqrt{2}\, dB_t$: Brownian noise to explore and escape.

Why this SDE targets $\pi(x) \propto e^{-V(x)}$

The Langevin diffusion is designed so that its **stationary distribution** is

$$\pi(x) \propto e^{-V(x)}.$$

Mental model

A particle moves on the landscape $V(x)$:

- drift pulls it toward low V,
- noise shakes it around.

At equilibrium, the time spent in a region is proportional to $e^{-V(x)}$.

Important: this is sampling, not optimization. Optimization removes noise and collapses to $\arg\min V$.

Why is it called "log-concave sampling"?

A distribution π is **log-concave** if $\log \pi(x)$ is concave.
If

$$\pi(x) = \frac{e^{-V(x)}}{Z},$$

then

$$\log \pi(x) = -V(x) - \log Z.$$

So:

$$\log \pi \text{ concave} \quad \Longleftrightarrow \quad V \text{ convex.}$$

Why this matters

Convex V typically implies:

- no separated modes (no severe multimodal trapping),
- well-behaved geometry,
- faster mixing / stronger theory guarantees.

Continuous-time Langevin diffusion

We construct a continuous-time process whose long-run law is the target

$$\pi(x) \propto e^{-V(x)}.$$

Langevin SDE

$$dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t,$$

where B_t is standard Brownian motion in $\mathbb{R}^d$.

- Drift term $-\nabla V$: pulls toward low-energy (high-probability) regions.
- Noise term $\sqrt{2}\,dB_t$: keeps exploration; prevents collapse to $\arg\min V$.

Theorem: invariant measure & convergence to π

Assume $V : \mathbb{R}^d \to \mathbb{R}$ is m-**strongly convex** and L-**smooth**:

$$mI_d \preceq \nabla^2 V(x) \preceq LI_d \quad \forall x.$$

Equivalently, $\pi(x) \propto e^{-V(x)}$ is **strongly log-concave**.

Theorem (stationarity + exponential convergence)

Let $(X_t)_{t\geq 0}$ solve the Langevin SDE. Then:

- The density $\pi(x) = \frac{e^{-V(x)}}{Z}$ is an **invariant** (stationary) distribution.
- Moreover, the law of X_t converges to π exponentially fast. For example, in KL:

$$\mathrm{KL}\big(\mathcal{L}(X_t) \,\|\, \pi\big) \;\leq\; e^{-2mt}\,\mathrm{KL}\big(\mathcal{L}(X_0) \,\|\, \pi\big).$$

(Similar exponential decay holds in W_2 and other metrics under these assumptions.)

Proof sketch (why it converges to π)

Let ρ_t be the density of X_t.

1) Invariance via Fokker–Planck

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla V) + \Delta \rho_t.$$

Plug $\rho_\star(x) \propto e^{-V(x)}$:

$$\nabla \rho_\star = -\rho_\star \nabla V \quad \Rightarrow \quad \Delta \rho_\star = \nabla \cdot (-\rho_\star \nabla V),$$

so $\nabla \cdot (\rho_\star \nabla V) + \Delta \rho_\star = 0$ and $\rho_\star$ is stationary.

2) Exponential convergence (entropy dissipation)

$$\frac{d}{dt} \mathrm{KL}(\rho_t \| \pi) = -I(\rho_t \| \pi).$$

Strong log-concavity (m-convex V) $\Rightarrow$ log-Sobolev:

$$I(\rho_t \| \pi) \geq 2m \, \mathrm{KL}(\rho_t \| \pi) \quad \Rightarrow \quad \frac{d}{dt} \mathrm{KL}(\rho_t \| \pi) \leq -2m \, \mathrm{KL}(\rho_t \| \pi).$$

Grönwall:

$$\mathrm{KL}(\rho_t \| \pi) \leq e^{-2mt} \mathrm{KL}(\rho_0 \| \pi).$$

Bayesian inference example (why sampling shows up)

We observe data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ and infer parameters $\theta \in \mathbb{R}^d$.

Posterior = prior × likelihood (up to normalization)

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)\, p(\theta).$$

- What we need are **integrals** under the posterior:

$$\mathbb{E}[f(\theta) \mid \mathcal{D}] = \int f(\theta)\, p(\theta \mid \mathcal{D})\, d\theta.$$

- Typical targets: posterior mean $\mathbb{E}[\theta \mid \mathcal{D}]$, uncertainty, and posterior predictive.
- Problem: in high-d, the posterior is rarely a closed-form distribution we can sample from directly.

Concrete model: Bayesian logistic regression $\Rightarrow$ log-concave posterior

Binary labels $y_i \in \{0, 1\}$, features $x_i \in \mathbb{R}^d$. Let $\sigma(t) = \frac{1}{1+e^{-t}}$ and

$$\Pr(y_i = 1 \mid x_i, \theta) = \sigma(x_i^\top \theta).$$

Gaussian prior:

$$\theta \sim \mathcal{N}(0, \alpha^{-1} I_d).$$

Energy form

$$p(\theta \mid \mathcal{D}) = \frac{e^{-V(\theta)}}{Z}, \quad V(\theta) = \sum_{i=1}^{n} \Big(\log(1 + e^{x_i^\top \theta}) - y_i x_i^\top \theta \Big) + \frac{\alpha}{2} \|\theta\|^2.$$

- Low $V(\theta) \Rightarrow$ high posterior density.
- Here V is convex $\Rightarrow$ the posterior is **log-concave**.

The problem with the continuous-time equation

The Langevin SDE

$$dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t$$

is a **continuous-time** stochastic process.

- It is defined for all real times $t \geq 0$.
- We cannot simulate B_t exactly on a computer.
  - Brownian motion paths are extremely rough: with probability 1, B_t is **nowhere differentiable**.
  - Indeed,

$$\frac{B_{t+h} - B_t}{h} \sim \mathcal{N}\left(0, \frac{1}{h}\right),$$

so as $h \to 0$ the variance blows up and no finite slope can exist.

Key limitation:

We cannot generate exact samples from the SDE itself.

Consequence

To obtain samples in practice, we must **discretize time**.

Discretizing Langevin: Euler–Maruyama (ULA)

The Langevin SDE

$$dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t$$

is continuous in time and cannot be simulated exactly.

Fix a step size $h > 0$ and observe the process at discrete times

$$t_k = kh, \qquad X_k := X_{t_k}.$$

Integrating the SDE over $[t_k, t_{k+1}]$ gives

$$X_{k+1} - X_k = -\int_{t_k}^{t_{k+1}} \nabla V(X_t)\,dt + \sqrt{2}\,(B_{t_{k+1}} - B_{t_k}).$$

Discretizing Langevin: Euler–Maruyama (ULA)

- Approximate by freezing it at X_k:

$$\int_{t_k}^{t_{k+1}} \nabla V(X_t)\, dt \;\approx\; h\,\nabla V(X_k).$$

- Brownian increments satisfy $B_{t_{k+1}} - B_{t_k} \sim \mathcal{N}(0, hI_d)$, which explains the $\sqrt{h}$ scaling.

This yields the update

$$X_{k+1} = X_k - h\,\nabla V(X_k) + \sqrt{2h}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, I_d),$$

known as the **Unadjusted Langevin Algorithm (ULA)**.

Does discretization preserve the target distribution?

Let π_h denote the stationary distribution of the discretized chain (ULA):

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_k.$$

Question

Is $\pi_h = \pi$?

Bad news: in general, **no**. Time discretization changes the invariant distribution.

Easiest example: quadratic potential

Take

$$V(x) = \tfrac{1}{2}\|x\|^2 \qquad \Rightarrow \qquad \pi = \mathcal{N}(0, I_d).$$

Then ULA becomes the linear recursion

$$X_{k+1} = (1-h)X_k + \sqrt{2h}\,\xi_k.$$

This is a process with stationary distribution

$$\pi_h = \mathcal{N}\Big(0,\ \tfrac{2}{2-h}\, I_d\Big), \qquad h < 2.$$

Conclusion: Even in the simplest Gaussian case,

$$\pi_h \neq \pi \quad \text{for any } h > 0.$$

The bias vanishes only in the limit $h \to 0$.

If $\pi_h \neq \pi$, are they at least close?

Although discretization changes the invariant distribution, the bias can be **quantified and controlled**.

- Let π be the invariant law of the Langevin diffusion.
- Let π_h be the invariant law of ULA with step size h.

Question: As $h \to 0$, does $\pi_h \to \pi$?

Answer: yes, under standard regularity assumptions.

Assume V is:

- m-strongly convex, and
- L-smooth (Lipschitz gradient).

Then for sufficiently small h,

$$W_2(\pi_h, \pi) \;\leq\; C\,h,$$

where C depends on (m, L, d).

- Smaller $h \Rightarrow$ smaller stationary bias.
- Smaller $h \Rightarrow$ stronger correlation and slower mixing.

Why is π_h close to π in W_2? (local view)

Let $Q_h(x, \cdot)$ denote one exact Langevin step of duration h, and $P_h(x, \cdot)$ one ULA step, both starting from x.

Couple the two steps using the same Brownian increment:

$$X_h = x - \int_0^h \nabla V(X_t)\, dt + \sqrt{2}(B_h - B_0),$$

$$\tilde{X}_h = x - h\nabla V(x) + \sqrt{2h}\,\xi, \qquad \xi = \frac{B_h - B_0}{\sqrt{h}}.$$

Then the difference is purely a drift error:

$$X_h - \tilde{X}_h = -\int_0^h \big(\nabla V(X_t) - \nabla V(x)\big)\, dt.$$

Using L-smoothness and $\mathbb{E}\|X_t - x\| = O(\sqrt{t})$,

$$\mathbb{E}\|X_h - \tilde{X}_h\| \;\le\; \int_0^h L\,\mathbb{E}\|X_t - x\|\, dt = O(h^{3/2}).$$

Thus, uniformly in x,

Where does the W_2 contraction come from?

The contraction is a **geometric consequence of strong convexity** of V.

- If V is m-strongly convex, then its gradient is *monotone*:

$$\langle \nabla V(x) - \nabla V(y),\, x - y \rangle \;\geq\; m\|x - y\|^2.$$

- Consider two ULA chains started from x, y and driven by the *same* noise (synchronous coupling):

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_k,$$
$$Y_{k+1} = Y_k - h\nabla V(Y_k) + \sqrt{2h}\,\xi_k.$$

- The noise cancels in the difference:

$$X_{k+1} - Y_{k+1} = (X_k - Y_k) - h\big(\nabla V(X_k) - \nabla V(Y_k)\big).$$

- Using strong convexity and L-smoothness,

$$\|X_{k+1} - Y_{k+1}\|^2 \leq (1 - 2mh + h^2L^2)\,\|X_k - Y_k\|^2.$$

For sufficiently small h,

$$\|X_{k+1} - Y_{k+1}\| \leq (1 - mh)\,\|X_k - Y_k\|.$$

Why is π_h close to π in W_2? (global view)

$$W_2(\mu P_h, \nu P_h) \;\leq\; (1 - mh)\, W_2(\mu, \nu).$$

Now compare invariant measures:

$$W_2(\pi, \pi_h) = W_2(\pi Q_h, \pi_h P_h).$$

Insert πP_h and apply triangle inequality:

$$W_2(\pi, \pi_h) \leq W_2(\pi Q_h, \pi P_h) + W_2(\pi P_h, \pi_h P_h).$$

Using:

- one-step bound: $W_2(\pi Q_h, \pi P_h) = O(h^{3/2})$,
- contraction: $W_2(\pi P_h, \pi_h P_h) \leq (1 - mh) W_2(\pi, \pi_h)$,

we obtain

$$mh\, W_2(\pi, \pi_h) \;\leq\; O(h^{3/2}).$$

$$\Longrightarrow \qquad W_2(\pi_h, \pi) = O(\sqrt{h}).$$

Interpretation: Local discretization errors are small and are damped by global contraction.

Quadratic example: compute $W_2(\pi_h, \pi)$ explicitly

Take $V(x) = \frac{1}{2}\|x\|^2$, so the target is

$$\pi = \mathcal{N}(0, I_d).$$

ULA gives

$$X_{k+1} = (1-h)X_k + \sqrt{2h}\,\xi_k \qquad \Rightarrow \qquad \pi_h = \mathcal{N}\Big(0,\ aI_d\Big), \quad a = \frac{2}{2-h}, \quad h < 2.$$

For centered Gaussians, the Wasserstein–2 distance satisfies

$$W_2^2\big(\mathcal{N}(0,\Sigma_1), \mathcal{N}(0,\Sigma_2)\big) = \mathrm{tr}(\Sigma_1) + \mathrm{tr}(\Sigma_2) - 2\,\mathrm{tr}\Big((\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2})^{1/2}\Big).$$

With $\Sigma_1 = I_d$ and $\Sigma_2 = aI_d$, this simplifies to

$$W_2(\pi_h, \pi) = \sqrt{d}\left(\sqrt{\frac{2}{2-h}} - 1\right) = \sqrt{d}\left(\frac{1}{\sqrt{1-h/2}} - 1\right).$$

Is this good news? Yes: for small h,

$$\frac{1}{\sqrt{1-h/2}} - 1 = \frac{h}{4} + O(h^2) \quad \Rightarrow \quad W_2(\pi_h, \pi) = \frac{\sqrt{d}}{4}\,h + O(h^2).$$

What ULA guarantees (and what it costs)

There are two fundamentally different issues:

- **Initialization bias:** the chain starts from $X_0 \sim \mu_0 \neq \pi$.
- **Discretization bias:** even at stationarity, ULA targets $\pi_h \neq \pi$ in general.

As a result, ULA analysis naturally splits into:

- existence of a stationary distribution π_h,
- mixing / burn-in to reach π_h,
- bias from $\pi_h \neq \pi$,
- Monte Carlo error from finite-time averaging.

Key theme: Accuracy always trades off against computational cost.

What does the quadratic example teach us?

The Gaussian example is completely explicit, and it confirms the general theory.

From the exact computation,

$$W_2(\pi_h, \pi) = \sqrt{d}\left(\sqrt{\frac{2}{2-h}} - 1\right) = \frac{\sqrt{d}}{4}\, h + O(h^2).$$

- The discretization bias in W_2 is **linear in** h.
- The bias grows like $\sqrt{d}$ with the dimension.
- There is no cancellation: $\pi_h \neq \pi$ for every $h > 0$.

Good news: The bias vanishes at the optimal first-order rate as $h \to 0$.

Bad news: Making h small increases correlation and computational cost.

Mixing / burn-in: getting close to π_h

Even before discretization bias matters, the chain must **forget its initialization**.

Let $X_0 \sim \mu_0$ and let π_h be the stationary distribution of ULA.

- Mixing (or burn-in) asks: how large must k be so that

$$d\big(\mathcal{L}(X_k), \pi_h\big) \le \varepsilon,$$

  for a chosen distance $d(\cdot,\cdot)$ (e.g. W_2, TV)?

Diffusion-time heuristic:

- One ULA step corresponds to time $\approx h$ of the continuous diffusion.
- Reaching a fixed mixing time T_{mix} typically requires

$$k \;\approx\; \frac{T_{\mathrm{mix}}}{h}.$$

In strongly convex settings, one often obtains bounds of the form

$$k(\varepsilon) = O\left(\frac{1}{h}\log\frac{1}{\varepsilon}\right),$$

up to problem-dependent constants.

Monte Carlo error: time averages

After burn-in, we estimate expectations by time averages along the chain.

Let m be a burn-in time and define

$$\hat{I}_N := \frac{1}{N}\sum_{k=1}^{N} f(X_{m+k}).$$

Ergodic theorem (LLN for Markov chains):

$$\hat{I}_N \xrightarrow[N\to\infty]{} \mathbb{E}_{\pi_h}[f].$$

under standard regularity assumptions,

$$\sqrt{N}\left(\hat{I}_N - \mathbb{E}_{\pi_h}[f]\right) \;\Rightarrow\; \mathcal{N}(0, \sigma_f^2).$$

The variance is governed by the **effective sample size**

$$N_{\text{eff}} := \frac{N}{\tau_{\text{int}}}, \qquad \tau_{\text{int}} := 1 + 2\sum_{\ell\geq 1}\rho_f(\ell),$$

where $\rho_f(\ell)$ are autocorrelations of $f(X_k)$.

Full error decomposition and final takeaways

After burn-in, the ULA estimator

$$\hat{I}_N := \frac{1}{N}\sum_{k=1}^{N} f(X_{m+k})$$

approximates $\mathbb{E}_\pi[f]$ with three distinct error sources.

$$\left|\hat{I}_N - \mathbb{E}_\pi[f]\right| \leq \underbrace{\left|\mathbb{E}_{\pi_h}[f] - \mathbb{E}_\pi[f]\right|}_{\text{discretization bias } \sim O(h)} + \underbrace{\left|\hat{I}_N - \mathbb{E}_{\pi_h}[f]\right|}_{\text{sampling error } \sim O(1/\sqrt{N_{\mathrm{eff}}})} + \underbrace{(\text{residual init. bias})}_{\text{small if } m \text{ large}}.$$

Accuracy–cost tradeoff:

- Smaller h reduces discretization bias,
- but increases correlation and burn-in cost,
- so $N_{\mathrm{eff}} \ll N$ when h is very small.

A common balancing strategy to reach error $\lesssim \varepsilon$ is

$$h \asymp \varepsilon, \qquad N_{\mathrm{eff}} \asymp \varepsilon^{-2},$$

leading to **polynomial cost** in $1/\varepsilon$.

Removing discretization bias: Metropolis correction (MALA)

MALA (Metropolis-Adjusted Langevin Algorithm):

- Propose a Langevin step (same as ULA):

$$Y = X - h\nabla V(X) + \sqrt{2h}\,\xi, \qquad \xi \sim \mathcal{N}(0, I_d).$$

- Accept Y with probability $\alpha(X, Y)$ chosen so that the chain is **reversible w.r.t.** π (detailed balance).
- Otherwise, stay at X (reject).

Key benefit: the stationary distribution is **exactly** the target:

$$\pi_{\text{MALA}} = \pi,$$

so there is **no discretization bias at stationarity**.

After burn-in, the remaining error decomposition becomes

$$\left|\hat{I}_N - \mathbb{E}_\pi[f]\right| \leq \underbrace{\left|\hat{I}_N - \mathbb{E}_\pi[f]\right|}_{\text{sampling error } \sim O\left(\frac{1}{\sqrt{N_{\text{eff}}}}\right)} + \underbrace{(\text{residual init. bias})}_{\text{small if } m \text{ large}}.$$

Exact sampling idea: rejection sampling

Rejection sampling is the simplest example of an **exact sampling method**.

Setup:

- Target distribution π (known up to normalization $\tilde{\pi} \propto \pi$).
- Proposal distribution μ, easy to sample from.
- An unnormalized density $\tilde{\mu}$ such that

$$\tilde{\mu}(x) \;\geq\; \tilde{\pi}(x) \quad \text{for all } x.$$

Algorithm (repeat until acceptance):

- Draw $X \sim \mu$.
- Accept X with probability $\tilde{\pi}(X)/\tilde{\mu}(X)$.

Key idea: Reject samples in regions where the proposal overshoots the target.

Rejection sampling: what it guarantees

Rejection sampling has two fundamental and complementary properties.

Fact 1 (Exactness): The accepted sample is drawn **exactly** from the target distribution π. There is *no bias*.

Fact 2 (Random cost): The number of proposals until acceptance is geometrically distributed, with

$$\mathbb{E}[\text{number of proposals}] = \frac{Z_\mu}{Z_\pi}, \qquad Z_\mu = \int \tilde{\mu}, \quad Z_\pi = \int \tilde{\pi}.$$

Interpretation:

- If $\tilde{\mu} \approx \tilde{\pi}$, acceptance is frequent.
- If $\tilde{\mu} \gg \tilde{\pi}$, the method becomes very inefficient.

Why rejection sampling is exact (proof sketch)

Let $(X_i)_{i\geq 1}$ be i.i.d. samples from μ and $(U_i)_{i\geq 1}$ be i.i.d. $\mathrm{Unif}[0,1]$, independent of (X_i).

The algorithm accepts the first index $n+1$ such that

$$U_{n+1} \leq \frac{\tilde{\pi}(X_{n+1})}{\tilde{\mu}(X_{n+1})}, \quad \text{and rejects all previous samples.}$$

For any measurable set A,

$$\begin{aligned}
\Pr(X \in A) &= \sum_{n=0}^{\infty} \Pr\Big(X_{n+1} \in A,\ U_{n+1} \leq \tfrac{\tilde{\pi}(X_{n+1})}{\tilde{\mu}(X_{n+1})},\ \text{first } n \text{ rejected}\Big) \\
&= \sum_{n=0}^{\infty} \Pr\Big(X_{n+1} \in A,\ U_{n+1} \leq \tfrac{\tilde{\pi}(X_{n+1})}{\tilde{\mu}(X_{n+1})}\Big) \Pr\Big(U_1 > \tfrac{\tilde{\pi}(X_1)}{\tilde{\mu}(X_1)}\Big)^n.
\end{aligned}$$

Using independence and integration,

$$\Pr(X \in A) = \left(\frac{Z_\pi}{Z_\mu}\right)\pi(A) \sum_{n=0}^{\infty} \left(1 - \frac{Z_\pi}{Z_\mu}\right)^n = \pi(A).$$

MALA: Metropolis-Adjusted Langevin Algorithm

MALA uses the same Langevin *proposal* as ULA, but adds an accept/reject step.

Proposal (Langevin step): from the current state x, propose

$$Y = x - h\nabla U(x) + \sqrt{2h}\,\xi, \qquad \xi \sim \mathcal{N}(0, I_d).$$

Acceptance step: accept the proposal Y with probability

$$A(x,y) = \min\left\{1,\ \frac{\pi(y)\ \exp\big(-\frac{1}{4h}\|x-y+h\nabla U(y)\|^2\big)}{\pi(x)\ \exp\big(-\frac{1}{4h}\|y-x+h\nabla U(x)\|^2\big)}\right\}.$$

If accepted, set $X_{k+1} = Y$; otherwise, set $X_{k+1} = X_k$.

Metropolis–Hastings: transition rule and reversibility

Acceptance probability (MH):

$$A(x,y) = \min\left\{1,\ \frac{\pi(y)\,q(y \to x)}{\pi(x)\,q(x \to y)}\right\}.$$

Transition law (move-or-stay): for any measurable set B,

$$\Pr(X_{k+1} \in B \mid X_k = x) = \int_B q(x \to y)\,A(x,y)\,dy \ + \ \left(1 - \int q(x \to y)\,A(x,y)\,dy\right)\mathbf{1}_{\{x \in B\}}.$$

(First term = accepted moves, second term = rejection probability $\Rightarrow$ stay at x.)

Reversibility (detailed balance): it suffices to check that for $x \neq y$,

$$\pi(x)\,q(x \to y)\,A(x,y) = \pi(y)\,q(y \to x)\,A(y,x).$$

Proof (one line):

$$\pi(x)q(x \to y)A(x,y) = \pi(x)q(x \to y)\min\left\{1, \frac{\pi(y)q(y \to x)}{\pi(x)q(x \to y)}\right\} = \min\{\pi(x)q(x \to y),\ \pi(y)q(y \to x)\},$$

which is symmetric in (x,y), hence equals $\pi(y)q(y \to x)A(y,x)$.

Cold-start MALA: goal and reduction

We consider MALA targeting

$$\pi(x) \propto e^{-V(x)}, \qquad \alpha I \preceq \nabla^2 V(x) \preceq \beta I, \qquad \kappa := \beta/\alpha.$$

Cold start:

$$X_0 \sim \mu_0 := \mathcal{N}(0, \beta^{-1} I_d), \qquad \mu_N := \mathcal{L}(X_N).$$

Goal: for a tuned step size h, show

$$\chi^2(\mu_N \| \pi) \;\leq\; \varepsilon \quad \text{after} \quad N = \widetilde{O}\big(\kappa d \ \mathrm{polylog}(1/\varepsilon)\big).$$

Key reduction: Because MALA is reversible w.r.t. π, its Markov operator P is self-adjoint in $L^2(\pi)$, yielding geometric contraction:

$$\chi^2(\mu_0 P^N \| \pi) \leq (1 - \lambda(P))^N \, \chi^2(\mu_0 \| \pi).$$

Thus it suffices to bound:

- the initial divergence $\chi^2(\mu_0 \| \pi)$,
- the spectral gap $\lambda(P)$ of MALA.

Notation

- **Target:** $\pi(dx) \propto e^{-V(x)}\,dx$ on $\mathbb{R}^d$, with

$$\mu I \preceq \nabla^2 V(x) \preceq LI \quad \forall x.$$

- **LSM (step size h):**

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_{k+1}, \qquad \xi_{k+1} \sim \mathcal{N}(0, I_d).$$

- **Laws / measures:**
  - $\mathcal{L}(X_T)$: distribution (law) of X_T.
  - π_h: **invariant law** of the LSM Markov chain (i.e., if $X_0 \sim \pi_h$ then $X_k \sim \pi_h$ for all k).
  - δ_x: Dirac measure at x .
- **Occupation / empirical measure after burn-in T:**

$$\widehat{\mu}_{n,T} := \frac{1}{n} \sum_{k=T+1}^{T+n} \delta_{X_k}.$$

- **Wasserstein distances:** $W_1(\cdot,\cdot)$ and $W_2(\cdot,\cdot)$.
- **Gaussian smoothing:** $N_\sigma = \mathcal{N}(0, \sigma^2 I_d)$ and

$$W_1^{(\sigma)}(\mu,\nu) := W_1(\mu * N_\sigma,\ \nu * N_\sigma).$$

From $W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi)$ to Burn-in + Bias + Sampling

Step 1: Add/subtract π_h (triangle inequality).

$$W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi) \leq W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi_h) + W_1^{(\sigma)}(\pi_h, \pi).$$

Step 2: Compare smoothed W_1 to W_2 (since $W_1 \leq W_2$).

$$W_1^{(\sigma)}(\pi_h, \pi) \leq W_1(\pi_h, \pi) \leq W_2(\pi_h, \pi) \qquad \Rightarrow \qquad \boxed{\text{Discretization bias term}}.$$

Step 3: Add/subtract $\mathcal{L}(X_T)$ (to expose burn-in).

$$W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi_h) \leq W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) + W_1^{(\sigma)}(\mathcal{L}(X_T), \pi_h).$$

Step 4: Contractivity / mixing of ULA in W_2 under strong convexity.

$$W_2(\mathcal{L}(X_T), \pi_h) \leq (1 - \mu h)^{T/2}\, W_2(\delta_{x_0}, \pi_h),$$

and since $W_1^{(\sigma)} \leq W_1 \leq W_2$,

$$W_1^{(\sigma)}(\mathcal{L}(X_T), \pi_h) \leq (1 - \mu h)^{T/2}\, W_2(\delta_{x_0}, \pi_h) \qquad \Rightarrow \qquad \boxed{\text{Burn-in term}}.$$

What remains (the hard part):

$$\boxed{W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T))} \quad \Rightarrow \quad \boxed{\text{Sampling + dependence term (handled by concentration + covering)}}.$$

Proof of ULA contraction : synchronous coupling (1/2)

ULA dynamics:

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_{k+1}, \qquad \xi_{k+1} \sim \mathcal{N}(0, I_d).$$

Synchronous coupling: run two chains with the *same* noise ξ_{k+1}:

$$Y_{k+1} = Y_k - h\nabla V(Y_k) + \sqrt{2h}\,\xi_{k+1}.$$

Let $\Delta_k := X_k - Y_k$. Then noise cancels and

$$\Delta_{k+1} = \Delta_k - h\big(\nabla V(X_k) - \nabla V(Y_k)\big).$$

Square and use strong convexity + smoothness:

$$\|\Delta_{k+1}\|^2 = \|\Delta_k\|^2 - 2h\langle \Delta_k, \nabla V(X_k) - \nabla V(Y_k)\rangle + h^2\|\nabla V(X_k) - \nabla V(Y_k)\|^2.$$

If $\nabla^2 V \succeq \mu I$ and $\nabla^2 V \preceq LI$, then

$$\langle x - y, \nabla V(x) - \nabla V(y)\rangle \geq \mu\|x - y\|^2, \qquad \|\nabla V(x) - \nabla V(y)\| \leq L\|x - y\|.$$

Hence

$$\|\Delta_{k+1}\|^2 \leq (1 - 2\mu h + L^2 h^2)\,\|\Delta_k\|^2 \leq (1 - \mu h)\,\|\Delta_k\|^2 \quad \text{(e.g. if } h \leq 1/(2L)\text{)}.$$

Proof of ULA contraction: from coupling to W_2 (2/2)

Convert to W_2 and iterate:

$$W_2^2(\mathcal{L}(X_{k+1}), \mathcal{L}(Y_{k+1})) \leq \mathbb{E}\|\Delta_{k+1}\|^2 \leq (1-\mu h)\,\mathbb{E}\|\Delta_k\|^2$$

so

$$W_2(\mathcal{L}(X_T), \mathcal{L}(Y_T)) \leq (1-\mu h)^{T/2} W_2(\mathcal{L}(X_0), \mathcal{L}(Y_0)).$$

Set $X_0 = x_0$ and $Y_0 \sim \pi_h$ (invariant $\Rightarrow$ $Y_T \sim \pi_h$):

$$\boxed{W_2(\mathcal{L}(X_T), \pi_h) \leq (1-\mu h)^{T/2}\, W_2(\delta_{x_0}, \pi_h).}$$

Bounding $W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T))$: sup

Goal: control the *sampling/dependence* term

$$W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) = W_1\big(\widehat{\mu}_{n,T} * N_\sigma,\ \mathcal{L}(X_T) * N_\sigma\big).$$

Step 1 (Kantorovich–Rubinstein duality).

$$W_1(\alpha, \beta) = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \Big(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f]\Big).$$

Apply to $\alpha = \widehat{\mu}_{n,T} * N_\sigma$, $\beta = \mathcal{L}(X_T) * N_\sigma$:

$$W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left| \frac{1}{n} \sum_{k=T+1}^{T+n} (f * \varphi_\sigma)(X_k) - \mathbb{E}\,(f * \varphi_\sigma)(X_T) \right|.$$

Define the smoothed test class:

$$\mathcal{G}_\sigma := \{g = f * \varphi_\sigma :\ \|f\|_{\mathrm{Lip}} \le 1\}.$$

Then

$$\boxed{W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) = \sup_{g \in \mathcal{G}_\sigma} \left| \frac{1}{n} \sum_{k=T+1}^{T+n} g(X_k) - \mathbb{E}\, g(X_T) \right|.}$$

Bounding $W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T))$: covering

Step 2 (Covering / net idea). Fix $\varepsilon > 0$ and take an ε-net $\{g_1, \dots, g_N\}$ of $\mathcal{G}_\sigma$ in a chosen norm (e.g., $L_2(\pi_h)$ on a truncated ball):

$$N = N(\varepsilon, \mathcal{G}_\sigma, \|\cdot\|).$$

For any $g \in \mathcal{G}_\sigma$, pick g_j with $\|g - g_j\| \le \varepsilon$ and use triangle:

$$\left|\frac{1}{n}\sum g(X_k) - \mathbb{E}g(X_T)\right| \le \underbrace{\left|\frac{1}{n}\sum g_j(X_k) - \mathbb{E}g_j(X_T)\right|}_{\text{finite class}} + \underbrace{\left|\frac{1}{n}\sum (g - g_j)(X_k)\right|}_{\text{empirical approx}} + \underbrace{|\mathbb{E}(g - g_j)(X_T)|}_{\text{population approx}}.$$

Step 3 (finite class + union bound). If each g_j is bounded (after truncation) by $|g_j| \le b$ and the chain has pseudo spectral gap γ_{ps}, then concentration gives for each j:

$$\Pr\left(\left|\frac{1}{n}\sum g_j - \mathbb{E}g_j\right| > t\right) \lesssim \exp\Big(-c\, n\gamma_{\mathrm{ps}}\, \frac{t^2}{b^2}\Big),$$

and union bound over $j = 1, \dots, N$ yields:

$$\max_{j \le N}\left|\frac{1}{n}\sum g_j - \mathbb{E}g_j\right| \lesssim b\sqrt{\frac{\log N(\varepsilon, \mathcal{G}_\sigma, \|\cdot\|) + \log(1/\delta)}{n\gamma_{\mathrm{ps}}}}.$$

Sampling bound + discretization bias (compact)

(A) Sampling / dependence baseline (Step 4). Choose a net norm so both approximation terms are $\lesssim \varepsilon$ (e.g., L_∞ on a truncation ball, or $L_2(\pi_h)$ with a standard conversion). Optimize ε to get:

$$\boxed{W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) \;\lesssim\; b\sqrt{\frac{\log N(\varepsilon, \mathcal{G}_\sigma, \|\cdot\|) + \log(1/\delta)}{n_{\mathrm{eff}}}} \;+\; \varepsilon, \qquad n_{\mathrm{eff}} = n\gamma_{\mathrm{ps}}.}$$

(B) Discretization bias. ULA is Euler–Maruyama for the Langevin diffusion (invariant π), but ULA has invariant $\pi_h \neq \pi$:

$$dY_t = -\nabla V(Y_t)\,dt + \sqrt{2}\,dB_t, \qquad X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_{k+1}.$$

In the strongly log-concave regime ($\nabla^2 V \succeq \mu I$, $\nabla^2 V \preceq LI$, $h \leq 1/(2L)$), standard results give:

$$\boxed{W_2(\pi_h, \pi) \;\leq\; C(\mu, L)\,\sqrt{d\,h}.}$$

Intuition: why $d_{\log}(\sigma;\pi_h)$ shows up

complexity at scale σ:

$$\log N(\varepsilon, \mathcal{G}_\sigma, L_2(\pi_h)) \lesssim d_{\log}(\sigma;\pi_h)\,\log(1/\varepsilon), \qquad d_{\log}(\sigma;\pi_h) := \log\det\Big(I + \tfrac{1}{\sigma^2}\Sigma_h\Big).$$

Core idea:

- **W_1 = sup over 1-Lipschitz tests:** huge function class $\Rightarrow$ large covering numbers.
- **Gaussian smoothing ($* N_\sigma$) kills high frequencies:** in Fourier, $\widehat{g}(\omega) = \widehat{f}(\omega)\, e^{-\sigma^2\|\omega\|^2/2}$. Hence $\mathcal{G}_\sigma$ behaves like a *low-frequency / resolution-σ* class.
- **Complexity is measured in $L_2(\pi_h)$:** directions where π_h has small variance contribute less to $L_2(\pi_h)$ complexity (anisotropy matters).
- **Gaussian intuition:** if $X \sim \mathcal{N}(0,\Sigma)$, the relevant "modes" live in an ellipsoid set by $\Sigma + \sigma^2 I$; log-volume $\propto \log\det(\Sigma + \sigma^2 I)$. Factoring out $\sigma^2 I$ leaves

$$\log\det\Big(I + \tfrac{1}{\sigma^2}\Sigma\Big) = d_{\log}(\sigma).$$

Takeaway: $d_{\log}(\sigma;\pi_h)$ is the spectral "degrees-of-freedom" of π_h visible above noise level σ.

Gaussian target

Setting (Gaussian): $\pi = \mathcal{N}(0, \Sigma)$, and $\mathcal{G}_\sigma = \{g = f * \varphi_\sigma : \ \|f\|_{\mathrm{Lip}} \le 1\}$.

Claim:

$$\log N(\varepsilon, \mathcal{G}_\sigma, L_2(\pi)) \ \lesssim \ d_{\log}(\sigma; \pi) \ \log(1/\varepsilon), \qquad d_{\log}(\sigma; \pi) = \log\det\Big(I + \tfrac{1}{\sigma^2}\Sigma\Big).$$

Core steps:

- **Smoothing removes high frequencies:** in Fourier,

$$\widehat{g}(\omega) = \widehat{f}(\omega)\, e^{-\sigma^2\|\omega\|^2/2}.$$

- $L_2(\pi)$ **weights frequencies by Gaussian geometry:** for $\pi = \mathcal{N}(0, \Sigma)$, energy contributions are damped by $e^{-\omega^\top \Sigma\, \omega}$.
- **Effective frequency ellipsoid:** the relevant frequencies lie in

$$\Lambda(r) := \{\omega : \ \omega^\top(\Sigma + \sigma^2 I)\omega \le r^2\}.$$

  Tail outside $\Lambda(r)$ is $\le \varepsilon$ by choosing $r = r(\varepsilon)$.
- **Discretize inside** $\Lambda(r)$**:** approximate $\widehat{g}$ on a grid in $\Lambda(r)$ with resolution $\asymp \varepsilon \Rightarrow$ a finite ε-net in $L_2(\pi)$.
- **Count grid points:** $\log|\text{grid}| \ \approx \ \log \mathrm{Vol}(\Lambda(r)) \ + \ \log(1/\varepsilon)$, and

$$\log \mathrm{Vol}(\Lambda(r)) = \tfrac{1}{2}\log\det(\Sigma + \sigma^2 I) + O(d \log r).$$

Local quadratic approximation

Factoring out $\sigma^2 I$ yields the spectral term

$$\log\det(\Sigma + \sigma^2 I) = d\log\sigma^2 + \log\det\Big(I + \tfrac{1}{\sigma^2}\Sigma\Big).$$

Conclusion: complexity at resolution σ is governed by $d_{\log}(\sigma) = \log\det(I + \Sigma/\sigma^2)$.
Assume Hessian is Lipschitz:

$$\|\nabla^2 V(x) - \nabla^2 V(y)\|_{\mathrm{op}} \le M\|x - y\|.$$

Local quadratic (Taylor) approximation around the mode x:

$$V(x) = V(x^{)} + \tfrac{1}{2}(x - x^{)\top} H(x^{)}(x - x^{)} \;+\; R(x), \qquad H(x^{)} := \nabla^2 V(x^{)},$$

with cubic remainder bound

$$|R(x)| \;\le\; \frac{M}{6}\,\|x - x^{)}\|^3.$$

Perturbation idea: compare $\pi(x) \propto e^{-V(x)}$ to its Laplace Gaussian

$$g(x) \propto \exp\Big(-\tfrac{1}{2}(x - x^{)\top} H(x^{)}(x - x^{)}\Big) = \mathcal{N}(x^{)}, H(x^{)})^{-1}),$$

Local quadratic approximation

via the density ratio

$$\frac{\pi(x)}{g(x)} \propto \exp\big(-R(x)\big).$$

Blocking point (high dimension): typical radius. Under strong log-concavity ($\nabla^2 V \succeq \mu I$), typically

$$\|X - x^{\|}\| \approx \sqrt{d/\mu}.$$

Then the remainder exponent is typically

$$|R(X)| \lesssim \frac{M}{6}\Big(\frac{d}{\mu}\Big)^{3/2}.$$

Takeaway: unless $M(d/\mu)^{3/2} \ll 1$ (near-quadratic regime), π can be a *large* perturbation of the Gaussian Laplace approximation, so transferring Gaussian entropy bounds to general π is nontrivial.

When $d_{\log}(\sigma)$ improves $\log N$

Baseline: without exploiting geometry, covering numbers typically scale like

$$\log N(\varepsilon, \mathcal{G}_\sigma, \cdot) \;\sim\; d \cdot \log(1/\varepsilon) \quad \Rightarrow \quad W_1^{(\sigma)} \;\sim\; \sqrt{\frac{d}{n_{\text{eff}}}}.$$

Spectral refinement (Lemma 8):

$$\log N(\varepsilon, \mathcal{G}_\sigma, L_2(\pi_h)) \;\lesssim\; d_{\log}(\sigma; \pi_h)\, \log(1/\varepsilon), \quad d_{\log}(\sigma; \pi_h) = \sum_{i=1}^{d} \log\Big(1 + \frac{\lambda_i(\Sigma_h)}{\sigma^2}\Big).$$

Best case (anisotropic / low "active" dimension at scale σ):

- If many eigenvalues satisfy $\lambda_i(\Sigma_h) \ll \sigma^2$, then $\log(1 + \lambda_i/\sigma^2) \approx 0$.
- Then $d_{\log}(\sigma; \pi_h) \ll d$ and the sampling term improves to

$$W_1^{(\sigma)} \;\sim\; \sqrt{\frac{d_{\log}(\sigma; \pi_h)}{n_{\text{eff}}}} \quad \text{(strong improvement)}.$$

When $d_{\log}(\sigma)$ improves $\log N$

Worst case (isotropic / all directions equally relevant):

$$\lambda_i(\Sigma_h) \approx \lambda \ \forall i \quad \Rightarrow \quad d_{\log}(\sigma; \pi_h) \approx d \cdot \log\left(1 + \frac{\lambda}{\sigma^2}\right),$$

so the refinement reduces to essentially the original dimension-driven bound (up to a log factor).

Rule of thumb: $d_{\log}(\sigma)$ counts degrees-of-freedom *visible above noise level* σ.

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